

PromptEngineering

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Persona & Role-Play

- **The "Actor" Definition:** Assigning a specific role or expertise level to the model (e.g., "Act as a Senior Network Architect").
- **Align** the model with the rigorous "Gold Samples" of that specific field.
- A persona automatically sets the **tone, level of technicality, and preferred terminology** without needing a list of individual rules.
- **Triggers** specialized clusters of weights trained on high-quality, professional data (e.g., technical documentation, peer-reviewed journals).

Few Shot Injection

- Activating "**Pattern Mimicry**" strength of agent.
- Providing 2–5 examples of **(Input, Output)** pairs within the prompt before the final query.
- **In-Context Learning (ICL)**: Leverages the ability of LLMs to extrapolate patterns from the immediate context.
- **Format Enforcement**: The most effective way to ensure the model follows a specific structural constraint (e.g., "Always return a JSON object with keys 'ip_address' and 'status'").
- By seeing 3 correct examples, the **model's probability distribution** for the 4th response collapses around the established pattern.

Chain-of-Thought

- **Reasoning Activation:** Transitioning the model from "fast" intuitive pattern matching to "slow" analytical logical derivation.
- The Power of "**Let's think step-by-step**": A specific linguistic trigger that drastically shifts the probability distribution, forcing the model to generate intermediate reasoning tokens.
- **Intermediate State Representation:** Using the output window as a "scratchpad" to handle complex dependencies, multi-step math, or nested logic.
- **Technical Insight:** Each reasoning token acts as a new piece of context, effectively expanding the model's "working memory" for that specific inference.

Delimiters and Structure

- The Purpose of **Segmentation**: Using explicit visual markers (e.g., ###, "", ---, or XML tags like <context>) to delineate the boundaries between different prompt components.
- Preventing "**Attention Confusion**": Guides the model's self-attention mechanism to distinguish between the Command(the task) and the Payload (the data to be processed).
- **Structural Priming**: Providing a clear "start" signal for the response (e.g., "Result in JSON format: {" to lock the model into the desired output manifold immediately.
- **Defense** Against Prompt Injection: Isolates user-provided input from system instructions, making it harder for "untrusted" data to override the model's primary goal.

Negative Constraints

- Explicitly defining behaviors, styles, or information that the model **must avoid** during execution.
- **Effective Reframing:** Instead of "Don't be wordy," use "Be concise." Positive directives ("Only output the final answer") are often more stable than negative ones ("Do not include reasoning").
- Directly leverages the "**Harmlessness**" training from the SFT/RLHF phase, where models are taught to refuse specific categories of requests or styles.
- **Preventing Hallucination:** A critical negative constraint is the "I don't know" instruction.

Mapping

SFT	Prompt Engineering
Identity/Expert Samples	Persona/Role Play
Pattern Mimcry	Give few examples
Chain-of-Thought	Think step-by-step
Agentic Trajectories	Tool/MCP calls